Lecture-2

☆ Randomized Quicksort:

→ If we consider fix place of pivot, say last element always, someone can always come up with input on which Quicksort performs the worst i.e. n^2 comparisons.

→ So, the idea here is to randomly select pivot element from the array.

→ In each recursive call, the pivot is randomly selected from available data, so one cannot come up with the input like discussed earlier.

→ Let's say we have array A of 1000 elements

```
main()
{
Randomized_Quicksort(A, 1, 1000);
}

Randomized_Quicksort(A, p, r)
{
    if(p < r)
    {
        q ← Randomized_Partition(A, p, r);
        Randomized_Quicksort(A, p, q-1);
        Randomized_Quicksort(A, q+1, r);
    }
}
```

```
int Randomized_Partition(A, p, r)
{
    i ← Random(p, r); // Generate random
                      // index between p to r
    Exchange(A[i], A[r]); // Swap, So we can
                          // Proceed with Normal
                          // Quicksort
    return (Partition(A, p, r));
}

int Partition(A, p, r)
{
    int x = A[r];
    int i = p-1; // Initialization for lesser
    int j;       // element index

    for (j = p; j <= r-1; j++)
    {
        if (A[j] <= x) // Each is Compared to Pivot
        {
            i = i+1;
            Exchange(A[i], A[j]);
        }
    }
    Exchange(A[i+1], A[r]);
    return i+1;
}
```

Observation: Two elements S_i (With Rank=i) and S_j (With Rank=j) are compared if and only if

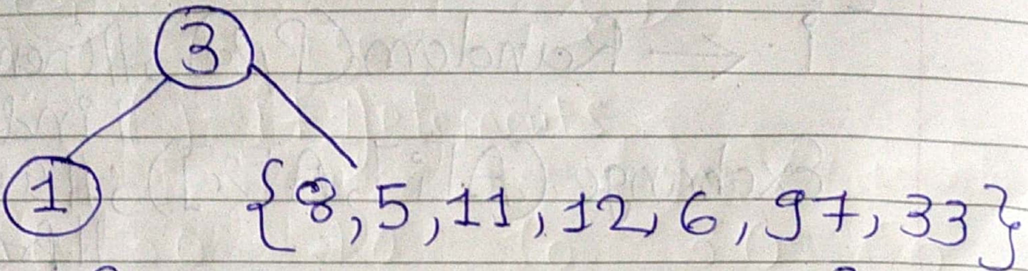
(i) Either S_i is pivot or

(ii) S_j is Pivot

$(i < j)$

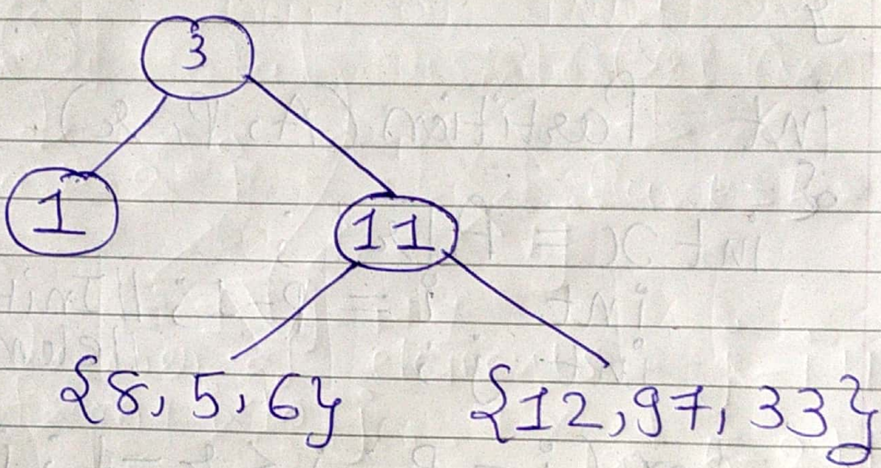
Ex: 1 8 5 3 11 12 6 97 33

→ Let's say Randomly Selected pivot is 3



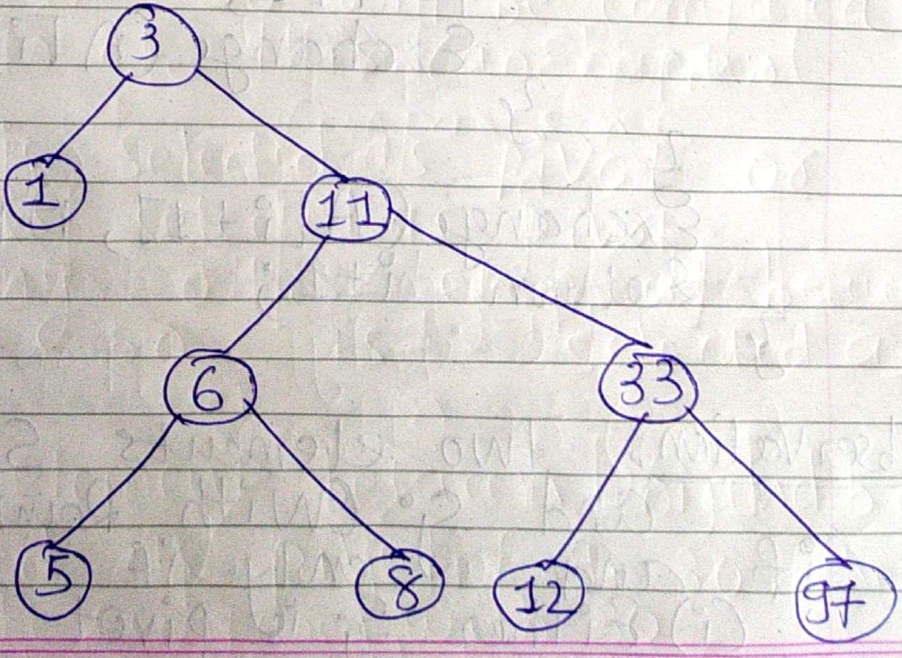
→ Now, for {8,5,11,12,6,97,33}, let's say 11 is selected as random pivot

∴



→ Now for {8,5,6}, let's say 6 is selected as pivot.

→ For {12,97,33}, 33 is selected as pivot.



→ Inorder Tree Traversal is

1, 3, 5, 6, 8, 11, 12, 33, 97.

→ Thus, Inorder traversal returns the sorted data.

☆ **Analysis:**

E.g. 20 50 40 10 30

Rank of 10 = 1
Rank of 20 = 2
Rank of 30 = 3
Rank of 40 = 4
Rank of 50 = 5

→ We denote element with rank = i as S_i
for E.g.

Rank of 10 = 1
$\uparrow$ $\uparrow$
S_i i

→ Recall the Observation:

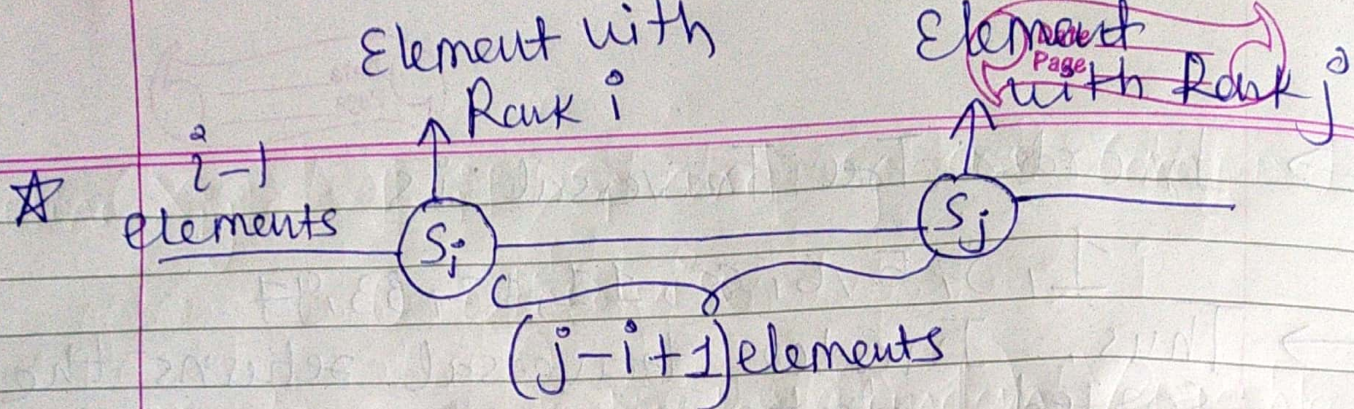
Assume $S_i \triangleq$ Element in the Array with Rank = i

$S_j \triangleq$ Element in the Array with Rank = j

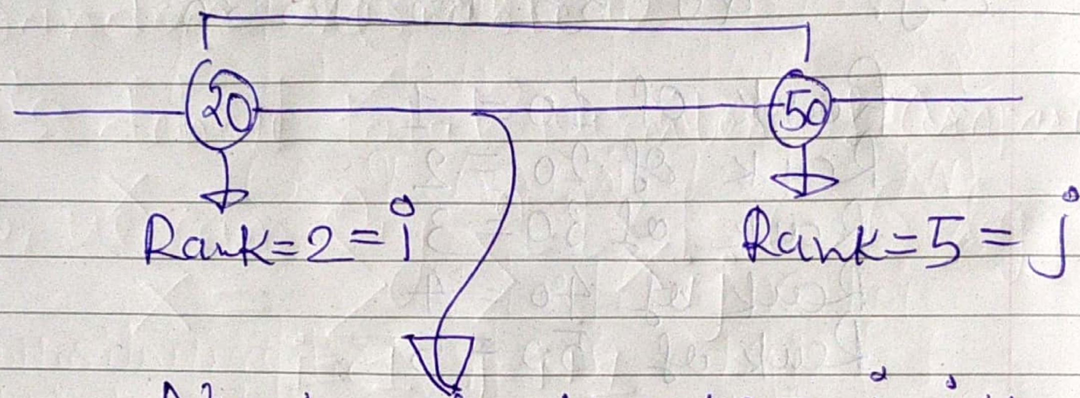
Assume $i < j$ i.e. $S_i < S_j$

Scanned with CamScanner

★



E.g. 20 50 40 10 30 60



Number of elements from i to j $= j-i+1$

$= 5-2+1$

$= 4$

i.e. <u>20, 30, 40, 50</u>

→ Now, 20 and 50 will be compared if either of this can happen

(i) 20 is Selected as pivot <u>or</u>

(ii) 50 is selected as pivot

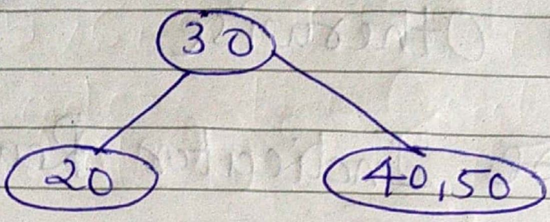
before any other element between 20 and 50 is selected as pivot.

→ If any element between 20 and 50 are used as pivot E.g. 30, then 20 will be compared with 30

50 will be Compared With 30

But 20 and 50 <u>Will not</u> be compared.

i.e. 20 and 50 will go in different partition.



Hence Conclusion

(i) Two elements S_i and S_j with rank i and j respectively will be compared iff either S_i or S_j is selected as pivot before any other element in between is selected as pivot.

(ii) If any other element is selected (between S_i and S_j) is selected as pivot then S_i and S_j will not be compared.

$Pr(S_i$ and S_j are compared$) = P_{ij}$

$$= \frac{2}{j-i+1}$$

$\Rightarrow$ i.e. either S_i or S_j is selected as pivot in the range from i to j.

- Total elements $= j-i+1$
- If any of these two is selected i.e. S_i and S_j then only S_i, S_j will be compared.

Scanned with CamScanner

Define $X_{ij} = 1$ if S_i and S_j are Compared
$= 0$ Otherwise

X_{ij}'s are Indicator Random Variable

$\therefore$ Total Number of Comparisons is random Variable.

Let's denote total Number of Comparisons as X.

$$X = \sum_{i=1}^{n} \sum_{j=i+1}^{n} X_{ij}$$

i.e. $X = X_{12} + X_{13} + \dots + X_{1n} +$
$X_{23} + X_{24} + \dots + X_{2n} +$
$\vdots \quad \dots \quad +$
$+ X_{(n-1)n}$

→ We want to find Expected Number of Comparisons.

$$\therefore E[X] = E\left[\sum_{i=1}^{n} \sum_{j=i+1}^{n} X_{ij}\right]$$

$$= \sum_{i=1}^{n} \sum_{j=i+1}^{n} E[X_{ij}]$$

$$= \sum_{i=1}^{n} \sum_{j=i+1}^{n} \Pr(S_i \text{ and } S_j \text{ are Compared})$$

Date ____
Page ____

$$= \sum_{i=1}^{n} \sum_{j=i+1}^{n} P_{ij}$$

$$= \sum_{i=1}^{n} \sum_{j=i+1}^{n} \frac{2}{j-i+1}$$

Let's say $j-i+1=k$

$\therefore$ when $j=i+1$, $i+1-i+1=k$

i.e. $\boxed{k=2}$

When $j=n$, $n-i+1=k$

i.e. $k=n-i+1$

$$\therefore E[X] = \sum_{i=1}^{n} \sum_{k=2}^{n-i+1} \frac{2}{k}$$

$$< \sum_{i=1}^{n} \sum_{k=2}^{n} \frac{2}{k}$$

Now, $\displaystyle\sum_{k=2}^{n} \frac{2}{k} = 2\sum_{k=2}^{n}\frac{1}{k}$

$$= 2\left(\frac{1}{2}+\frac{1}{3}+\dots+\frac{1}{n}\right)$$

$$= 2\left(1+\frac{1}{2}+\dots+\frac{1}{n}-1\right)$$

$$\approx 2(\ln n - 1)$$

Scanned with CamScanner

$\therefore E[X] \leq \sum_{i=1}^{n} \sum_{k=2}^{n} \frac{2}{k}$

$\simeq \sum_{i=1}^{n} 2(\ln n - 1)$

$= 2n(\ln n - 1)$

i.e. $E[X] \leq 2n\ln n - 2n$

So, we can say $E[X] = O(n \log n)$